

Competency Framework Examples

Sample Tasks

Sample tasks have been listed here for various combinations of component, activity and constraint for the three representative competency areas: software, systems and applications. Many, but not all the tasks are atomic. The tasks are not restricted to those that require CS or KA Core topics only.

Software Competency Area

Component	Activity	Tasks
Program	Design	• Design efficient data structures for a problem • Design test cases to determine if a program is functionally correct • Design an API for a service
Program	Develop	• Write a program that meets a given specification • Automate testing of new code under development • Develop a program that leverages libraries and APIs • Work in a team to effectively to solve a problem
Program	Document	• Document a program • Consistently format source code.
Program	Evaluate	• Evaluate an existing application (open source or proprietary) as a whole or partial solution for meeting a defined requirement
Program	Maintain	• Refactor a program. • Perform code review to evaluate the quality of code
Program	Humanize	• Ensure fair and equitable access in a program • Document the accountability, responsibility and liability an individual/company assumes when releasing a given service/software/product • Incorporate legal and ethical privacy requirements into a given service/software/product's development cycle • Convey the benefits of diverse development teams and user bases on the services/software/products the company provides, as well as the impacts that a lack of diversity can have on them
Program	Improve	• Debug a program
Program	Research	• Compute the running time of a program

		Formally prove the correctness of code
Algorithm	Design	Design an efficient algorithm for a problem
Algorithm	Document	Explain how an algorithm satisfies a set of requirements
Algorithm	Evaluate	Evaluate the efficiency of an algorithm
Algorithm	Maintain	Redesign an algorithm to improve a non-functional requirement
Algorithm	Humanize	Justify that an algorithm provides fair and equitable access to data
Algorithm	Research	- Prove the correctness of an algorithm - Compute the run time efficiency of an algorithm
Language / Paradigm	Design	Select a language/paradigm for an application
Language / Paradigm	Document	- Justify the choice of a paradigm/language for a program - Write a white paper to describe how a program is translated into machine code and executed - Write a white paper explaining how a program executes in an efficient manner with respect to memory and CPU utilization
Language / Paradigm	Evaluate	- Evaluate the appropriateness of a language/paradigm for an application - Explain the benefits and challenges of converting an application into parallel/distributed versions - Write a white paper explaining how a program effectively utilizes language features to make it safe and secure

Systems Competency Area

Component	Activity	Tasks
Processor	Design	Revise a specification to enable parallel processing without violating other essential properties or features
Processor	Develop	- Develop a version of CPU-based application to run on a hardware accelerator (GPU, TPU, NPU) - Implement a parallel/distributed component based on a known algorithm
Processor	Evaluate	Evaluate the performance-watt of a machine learning model deployed on an embedded device
Processor	Improve	Identify and repair a performance problem due to sequential bottlenecks
Storage	Evaluate	Assess the performance implications of cache memories in an application

		• Apply knowledge of operating systems to assess page faults in CPU-GPU memory management and their performance impact on an accelerated application
I/O	Design	• Design software modules for sensor hardware integration
I/O	Develop	• Develop a sensing-actuator robotics arm for an automated manufacturing cell • Develop a benchmarking software tool to assess the performance gain in removing I/O bottlenecks in code
Communication	Design	• Design a networking protocol. • Design software that enables safe communication between processes
Communication	Develop	• Develop a networked application • Deploy and securely operate a network of wireless sensors. • Develop software that enables safe communication between processes
Communication	Evaluate	• Evaluate the performance of a network, in specific latency, throughput, congestion, and various service levels
Communication	Maintain	• Defend a network from an ongoing distributed denial-of-service attack
Communication	Humanize	• Write a white paper to explain social, ethical, and professional issues governing the design and deployment of a networked system
Communication	Improve	• Identify failures in a datacenter network • Identify and repair a performance problem due to communication or data latency
Architecture	Develop	• Deploy a system in a cloud environment • Deploy an application component on a virtualized container
Architecture	Evaluate	• Find the performance bottleneck of a given system architecture
Data	Design	• Design how a new application's data will be stored
Data	Develop	• Create a database for a new application
Data	Maintain	• Get data back online after a disruption (e.g. power outage)
Data	Humanize	• Produce a white paper assessing the social and ethical implications of collecting and storing the data from a new (or existing) application • Assess the legal and ethical implications of collecting and using customer/user data

Data	Improve	• Improve a database application's performance (speed) • Modify a concurrent system to use a more scalable, reliable or available data store
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Applications Competency Area

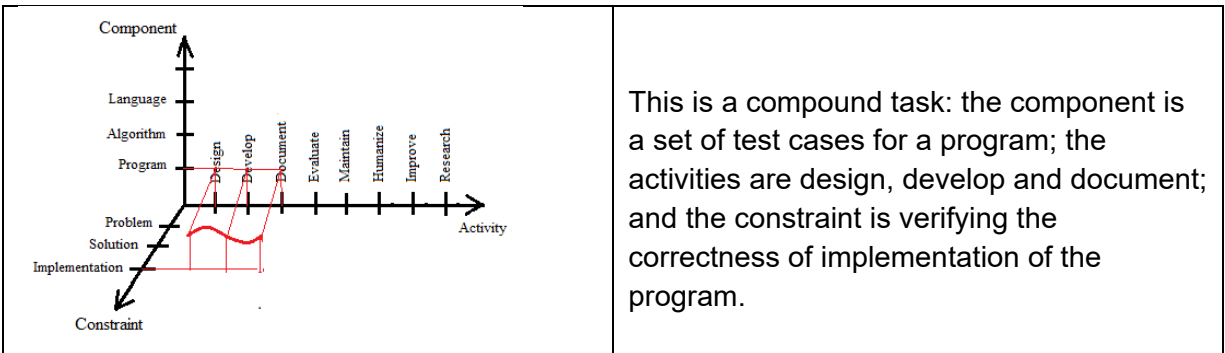
Component	Activity	Tasks
Input	Design	• Design a user interface for an application
Input	Develop	• Implement the user interface of an application
Input	Humanize	• Write a paper on the accessibility of a user interface
Computation	Design	• Specify the operators and partial-order planning graph to solve a logistics problem, showing all ordering constraints
Computation	Develop	• Implement an agent to play a two-player complete information board game • Write a program that uses Bayes rule to predict the probability of disease given the conditional probability table and a set of observations • Train a neural network to play a video game (e.g., Mario, Atari)
Computation	Evaluate	• Compare the performance of three supervised learning models on a dataset • Explain some of the pitfalls of deep generative models for image or text and how this can affect their use in an application
Computation	Humanize	• Write an essay on the effects of data set bias and how to mitigate them
Platform	Design	• Determine whether to develop an app as a native app or as a cross-platform app
Platform	Develop	• Create a mobile app that provides a consistent user experience across various devices, screen sizes, and operating systems • Develop a secure web interface for a business application
Platform	Evaluate	• Evaluate the usability and accessibility of an immersive system
Platform	Improve	• Optimize a dynamic web site for evolving business needs

Sample Competency Specifications

The following are some sample competency specifications for tasks that require various knowledge areas and skills. They have been listed under the three competency areas, viz., Software, Systems and Applications.

Software Competency Area

- **Task Software1:** Develop test cases to determine if a program is functionally correct.
- **Competency statement:** Develop test cases and test a given program.
- **Required knowledge:**
 - SDF-Practices
- **Required skills:** Develop
- **Desirable professional dispositions:** Meticulous, Persistent, Responsible



- **Task Software2:** Perform code review for a teammate.
- **Competency statement:** Communicate clearly and collaboratively to provide feedback to a teammate about a piece of code.
- **Required knowledge:**
 - SDF-Practices
 - SE-Teamwork
 - SE-Validation
- **Required skills:** Apply
- **Desirable professional dispositions:** Collaborative, Communicative, Meticulous

- **Task Software3:** Work on a team effectively.
- **Competency statement:** Focus on long-term team dynamics and communicate effectively.
- **Required knowledge:**
 - SE-Teamwork
- **Required skills:** Apply
- **Desirable professional dispositions:** Collaborative, Communicative, Proactive, Responsive

- **Task Software4:** Make an informed decision regarding which programming language/paradigm to select and use for a specific application.
- **Competency statement:** Apply knowledge of multiple programming paradigms, including their strengths and weaknesses relative to the application to be developed, and select an appropriate paradigm and programming language.

- **Required knowledge:**
 - FPL-OOP
 - FPL-Functional
 - FPL-Logic
 - FPL-Event-Driven
 - FPL-Types
 - FPL-Translation
 - FPL-Pragmatics
 - SPD-Embedded
 - FPL- Constructs
- **Required skills:** Explain, Evaluate
- **Desirable professional dispositions:** Inventive

- **Task Software5:** Effectively use a programming language's type system to develop safe and secure software.
- **Competency statement:** Apply knowledge of static type rules for a language to ensure an application is safe, secure, and correct.
- **Required knowledge:**
 - FPL-Types
- **Required skills:** Develop
- **Desirable professional dispositions:** Inventive, Meticulous

Systems Competency Area

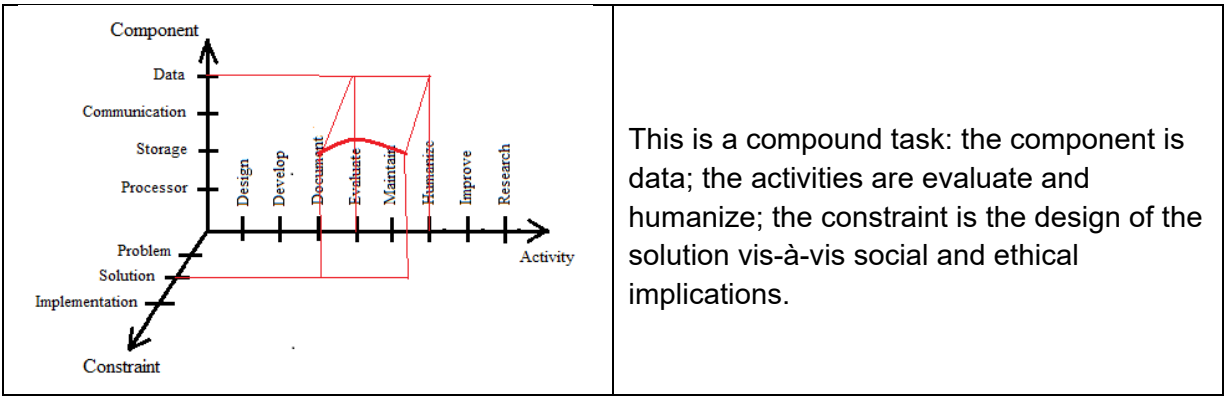
- **Task Systems1:** Develop a version of a CPU-based application to run on a hardware accelerator (GPU, TPU, NPU).
- **Competency statement:** Apply knowledge from systems design to accelerate an application code and evaluate the code speed-up.
- **Required knowledge:**
 - AR-Heterogeneity
 - PDC-Programs
 - SF-Design
- **Required skills:** Evaluate, Develop
- **Desirable professional dispositions:** Meticulous, Inventive

- **Task Systems2:** Produce a white paper assessing the social and ethical implications of collecting and storing data from a new (or existing) application.
- **Competency statement:** Identify the stakeholders and evaluate the potential long-term consequences of the collection and retention of data objects. Consider both potential harm from unintended data use and from data breaches.
- **Required knowledge:**
 - SEP-Context

- SEP-Ethical-Analysis
- SEP-Privacy
- SEP-Professional-Ethics
- SEP-Security
- SEP- DEIA
- DM-Data
- SEC-Foundations

● **Required skills:** Evaluate, Explain

● **Desirable professional dispositions:** Meticulous, Responsible, Proactive



- **Task Systems3:** Secure data from unauthorized access.
- **Competency statement:** Create database views to ensure data access is appropriately limited.
- **Required knowledge:**
 - DM-Data
 - DM-Relational
 - DM-Processing
 - SEP-Security
 - SEP-Professional-Ethics
 - SEP-Privacy
 - SEC-Foundations
- **Required skills:** Develop
- **Desirable professional dispositions:** Meticulous, Proactive

- **Task Systems4:** Create a database for a new application.
- **Competency statement:** Design the data storage needs (data modeling), assess the social and ethical implications for collecting and storing the data, determine how to store a new application's data (RDBMS vs NoSQL), and create the database, including appropriate indices.
- **Required knowledge:**
 - DM-Data

- DM-Core
- DM-Modeling
- DM-Relational
- DM-NoSQL
- DM-Internals
- SEP-Context
- SEP-Ethical-Analysis
- SEP-Privacy
- SEP-Professional-Ethics
- SEP-Security
- SEP- DEIA
- SEC-Foundations
- **Required skills:** Develop
- **Desirable professional dispositions:** Inventive, Meticulous, Responsible

- **Task Systems5:** Evaluate the performance of a network.
- **Competency statement:** Evaluate the latency, throughput, congestion, and various service levels of a network.
- **Required knowledge:**
 - NC-Applications
 - NC-Routing
- **Required skills:** Evaluate
- **Desirable professional dispositions:** Meticulous, Proactive

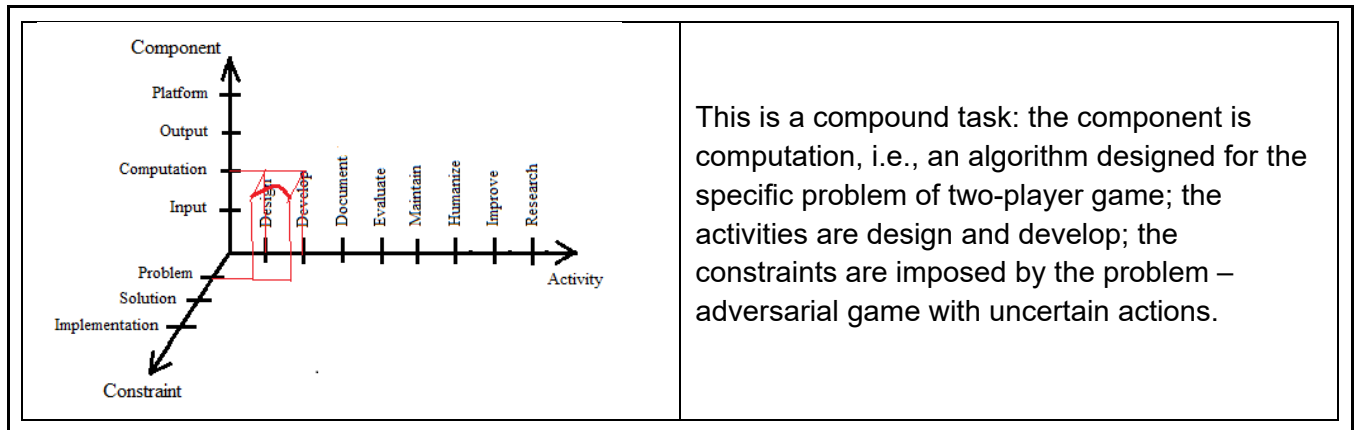
- **Task Systems6:** Deploy an application component on an operating system runtime/virtualized operating system/container.
- **Competency statement:** Identify and mitigate potential problems with deployment; automate setup of deployment environment; set up monitoring of component execution.
- **Required knowledge:**
 - OS-Purpose
 - OS-Principles
 - OS-Concurrency
 - OS-Scheduling
 - OS-Process
 - OS-Memory
 - OS-Protection
 - AR-Assembly
 - FPL-Scripting
- **Required skills:** Apply
- **Desirable professional dispositions:** Meticulous, Persistent, Proactive

- **Task Systems7:** Improve the performance of a sequential application or component by introducing parallelism.
- **Competency statement:** Evaluate how and when parallelism can improve (or not improve) performance well enough to identify opportunities, as well as implement them and measure the results.
- **Required knowledge:**
 - PDC-Evaluation
 - FPL-Parallel
- **Required skills:** Evaluate, Develop
- **Desirable professional dispositions:** Meticulous, Persistent, Proactive

- **Task Systems8:** Find the performance bottleneck of a given system.
- **Competency statement:** Given a system and its target deployment environment, find its performance bottleneck (e.g., memory, CPU, and networking) through analytical derivation or experimental study.
- **Required knowledge:**
 - SF-Performance
 - SF-Evaluation
 - SF-Design
 - SF-Overview
- **Required skills:** Apply, Evaluate, Develop
- **Desirable professional dispositions:** Meticulous, Persistent

Applications Competency Area

- **Task Applications1:** Implement an agent to make strategic decisions in a two-player adversarial game with uncertain actions (e.g., a board game, strategic stock purchasing).
- **Competency statement:** Use minimax with alpha-beta pruning, and possible chance nodes (expectiminimax), and heuristic move evaluation (at a particular depth) to solve a two-player zero-sum game.
- **Required knowledge:**
 - AI-Search
 - AI-KRR
- **Required skills:** Apply, Develop
- **Desirable professional dispositions:** Inventive, Persistent



- **Task Applications2:** Analyze tabular data (e.g., customer purchases) to identify trends and predict variables of interest.
- **Competency statement:** Use machine learning libraries, data preprocessing, training infrastructures, and evaluation methodologies to create a basic supervised learning pipeline.
- **Required knowledge:**
 - AI-ML
 - AI-SEP
- **Required skills:** Apply, Develop
- **Desirable professional dispositions:** Meticulous, Persistent, Responsible

- **Task Applications3:** Critique a deployed machine learning model in terms of potential bias and correct the issues.
- **Competency statement:** Understand, recognize, and evaluate issues of data set bias in AI, the types of bias, and algorithmic strategies for mitigation.
- **Required knowledge:**
 - AI-ML
 - AI-SEP
- **Required skills:** Explain
- **Desirable professional dispositions:** Responsible

- **Task Applications4:** Visualize a region's temperature record.
- **Competency statement:** Given weather data for a region, design and implement an animation depicting temperature changes over time.
- **Required knowledge:**
 - GIT-Fundamentals
 - GIT-Rendering
 - GIT-Visualization
 - HCI-Design
 - HCI-User

- **Required skills:** Apply, Evaluate, Develop
- **Desirable professional dispositions:** Inventive, Persistent

- **Task Applications5:** Evaluate and provide recommendations to improve a user-facing system.
- **Competency statement:** Apply knowledge of usability, accessibility, and inclusivity to evaluate a user-facing system.
- **Required knowledge:**
 - HCI-User
 - HCI-Accessibility
 - HCI-Evaluation
 - HCI-Design
 - HCI-SEP
- **Required skills:** Evaluate
- **Desirable professional dispositions:** Meticulous, Responsible

- **Task Applications6:** Determine the aspects of an implementation that require revision to support internationalization.
- **Competency statement:** Evaluate a system to identify culturally-relevant or language-relevant text, symbols, and patterns that may vary by locale.
- **Required knowledge:**
 - HCI-User
 - HCI-Accountability
 - HCI-Accessibility
 - HCI-Evaluation
- **Required skills:** Evaluate
- **Desirable professional dispositions:** Meticulous, Proactive, Responsible

- **Task Applications7:** Document the professional implications of a service/software/product for the company that produced it.
- **Competency statement:** Gather information regarding accountability, responsibility and liability assumed by a company when releasing a service/software/product and present it in a coherent and actionable manner.
- **Required knowledge:**
 - SEP-Professional-Ethics
 - SEP-IP
 - SEP-Privacy
 - SEP-Communication
 - SEP-DEIA
- **Required skills:** Explain
- **Desirable professional dispositions:** Meticulous, Proactive, Responsible

- **Task Applications8:** Determine whether to develop an app as a native app or as a cross-platform app.
- **Competency statement:** Understand performance and scalability issues, and evaluate different approaches and tools by carefully considering factors such as app requirements, target audience, time-to-market, and costs.
- **Required knowledge:**
 - SE-Tools
 - SPD-Common
 - SPD-Mobile
- **Required skills:** Explain
- **Desirable professional dispositions:** Inventive, Meticulous, Responsible

- **Task Applications9:** Build and optimize a secure web page for evolving business needs using a variety of *appropriate* programming languages.
- **Competency statement:** Evaluate potential security hazards and apply optimization techniques.
- **Required knowledge:**
 - AR-Performance-Energy
 - NC-Security
 - OS-Protection
 - SF-Security
 - SE-Design
 - SE-Tools
 - SPD-Common
 - SPD-Mobile
 - SEP-Privacy
- **Required skills:** Develop
- **Desirable professional dispositions:** Adaptable, Meticulous, Proactive, Responsible